

# Cal Methodology Stack — Navigation Index

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2026-05-20 Wednesday afternoon (v0.2 update 2026-05-21 Thursday morning per Phase 2 maturation)

## Cal Methodology Stack — Navigation Index

### Why this document exists

The Cal methodology stack grew rapidly through May 2026 to address specific discipline gaps as they emerged in K-audit chain operation, paper grade-pass work, and substrate-cognition + cosmology territory entry. As of 2026-05-20 EOD, the stack contains 8 standalone methodology documents + Appendix J in BST\_Referee\_Methodology.md + cross-references threading through the K-audit log and referee\_objections\_log.

For future-Cal arriving fresh, or for team CIs trying to find the right discipline reference for a specific question, this much methodology infrastructure can be hard to navigate. This index provides:

- 1. One-line description per document
- 2. When to use each
- 3. Cross-reference map
- 4. Categorization by discipline tier

This is navigation infrastructure, NOT new methodology content. Update this index when new methodology docs are filed.

### Quick-reference table

| Document                         | When to use                                                            | Tier            |
|----------------------------------|------------------------------------------------------------------------|-----------------|
| External_Survivability_Checklist | Before sending ANY external paper, letter, abstract, or Zenodo deposit | Pre-publication |

| Document                                                                      | When to use                                                                                | Tier                                |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------|
| Coincidence_Filter_Risk                                                       | Before drafting any X-fold-recurrence, multi-route convergence, cross-domain pattern claim | Negative filter                     |
| EXACT_vs_Mechanism_Distinction                                                | When EXACT identity claims are made — Mode 1 doesn't relax under precision                 | Negative filter                     |
| AuditChain_Quality_Pattern                                                    | When the audit chain produces convergent calibration patterns (M2C2)                       | Positive signal                     |
| Substrate_Cognition_Externalization                                           | Before ANY external mention of cognition-substrate territory                               | Pre-publication (highest severity)  |
| Claim_Level_Positive_Pattern                                                  | When claims involve overdetermined-form or cross-domain anchor clusters                    | Positive signal + Graph Forces spec |
| Trichotomy (Grace v0.1 + Cal Review)                                          | When catalog entries need ID / DER / PRED classification                                   | Catalog discipline                  |
| Methodology_Index (this document)                                             | When picking which discipline doc to apply                                                 | Navigation                          |
| Referee_Methodology Appendix J                                                | When ruling shape for K-audit candidates needs structural template                         | K-audit ruling shape                |
| Structurally_Verified_Tier_1 (Keeper per Cal #66)                             | When K-audit reaches BST-internal verification complete + alt-HSD pending                  | Audit-chain governance              |
| Phase1_Phase2_KAudit_Governance (Keeper per Cal #72)                          | When classifying K-audit as architectural-category vs INSTANCE-level chapter-category      | Audit-chain governance              |
| Dual-axis review rubric (per Cal #68 + #69 + #71 + #76, operational standing) | When grade-passing curriculum/paper artifacts at chapter-grade or paper-grade depth        | Curriculum/paper review             |

| Document                                                                                                          | When to use                                                                                                              | Tier                                                |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| BST TIER-1 FALSIFIER<br>SET external framing<br>(per Cal #74)                                                     | When external-presentation positioning of BST falsifiability claims (K87 + K90 + K91)                                    | External-presentation infrastructure                |
| RIGOROUSLY CLOSED tier supplement (Cal per #77, filed 2026-05-21 Thursday)                                        | When alt-HSD comparison demonstrates if-and-only-if mathematical-theorem distinguishability with EXACT BST-primary match | Audit-chain epistemic tier (11th methodology layer) |
| PCAP —<br>Pre-Specification<br>Cadence Acceleration<br>Pattern (Cal per #85, filed 2026-05-21 Thursday afternoon) | When recognizing execution-cycle compression in workstreams where methodology + pre-specs + reframing-insight converge   | Cadence pattern (16th methodology layer)            |

## Documents by category

Pre-publication discipline (external-facing material gate-pass)

1. **BST\_Methodology\_External\_Survivability\_Checklist.md** (Cal, 2026-05-17/18)

Apply before submitting any paper or sending any outreach letter externally. Names patterns that trigger crank-dismissal reflex (Eddington pattern, grand-claim register, statistical fog) and gives recommended replacements. Venue-specific (math journals vs physics journals vs popular-science). The 30-second outsider read is the operational test: would a senior referee categorize this paper as numerology in the first 30 seconds, or keep reading?

2. **BST\_Methodology\_Substrate\_Cognition\_External\_Register.md** (Cal, 2026-05-20)

Apply before ANY external communication touching on substrate-cognition territory (substrate-as-thinking, ideas-as-substrate-correlation, cognition-substrate hypothesis). MOST external-survivability-fraught territory in BST. Internal/external register separation with cognition-specific severity beyond standard Calibration #13. Nine red-flag trigger phrases catalogued; L2-hypothesis-with-falsifier-pairing standard mandatory; DEFAULT-DENY EXTERNAL standing. DOUBLE-LOCKED EXTERNAL tier (added 2026-05-20 afternoon) applies to compound dismissal-

pattern territory like cognition + cosmology (Penrose CCC + Penrose Orch-OR compound).

Negative-filter discipline (failure modes to clear)

3. **BST\_Methodology\_Coincidence\_Filter\_Risk.md** (Cal, 2026-05-17, Mode 6 thresholds added 2026-05-20)

Apply before drafting any claim of the form “BST integer X appears in N independent contexts,” “X-fold recurrence at integer Y,” “N-domain coverage at sub-percent precision,” “Classical theorem T produces a BST-decomposable integer set,” or “Cross-consistency network at N% pairwise.” Seven named failure modes:

1. Post-hoc clipping
2. Multi-decomposability
3. Search-space expansion
4. Ratio noise at low precision
5. Selection-effect on observables
6. Scan-protocol over/under-counting (Mode 6, Grace 2026-05-18; threshold formalization Cal 2026-05-20)
7. Classical-integer-set source claim without mechanism chain (Cal+Grace via K55, 2026-05-19)

Plus three diagnostic tests (Bonferroni/permutation null, forward predictions vs retrofit, null-model toy).

4. **BST\_Methodology\_EXACT\_vs\_Mechanism\_Distinction.md** (Cal, 2026-05-20 morning)

Apply when EXACT algebraic identity claims are made from BST primaries. Institutionalizes Calibration #13 (Keeper three-flag self-correction Tuesday EOD). Standing rule: EXACT identity verified at  $1e-14 \neq$  mechanism-forcing. Mode 1 (post-hoc form selection) does NOT relax under EXACT precision. Necessary but not sufficient: mechanism-derivation closure remains the D-tier gate.

Positive-signal discipline (structural-validation patterns)

5. **BST\_Methodology\_AuditChain\_Quality\_Patterns.md** (Cal + Keeper, 2026-05-20 morning)

Apply when the audit chain produces multi-CI convergent patterns. M2C2 (Multi-CI Convergent Calibration Pattern) first entry:  $\geq 2$  CIs applying independent discipline frameworks converging on same structural framing. Four documented instances by 2026-05-20 EOD: T2395 4+1 scope (Elie+Grace), Universal Q=126 cross-link (Elie+Lyra+Grace), K67 cascade extension (Lyra extending Elie), T2420 Four-Zone Vacuum Decomposition (Lyra+Elie). Audit-chain-level positive signal — NOT claim-level positive signal (those live in document 6).

6. **BST\_Methodology\_Claim\_Level\_Positive\_Patterns.md** (Cal, 2026-05-20 afternoon)

Apply when claims involve overdetermined-form clusters (Type 1 OFC) or cross-domain anchor clusters (Type 2 CDAC). Operational specification for Casey's Graph Forces candidate principle. Catalog tagging schema (OFC / CDAC / OFC+CDAC + provisional variants). CRITICAL CAUTION (added 2026-05-20 afternoon): Type 1+2 compound evidence does NOT multiply scalarly. Replace "N forms  $\times$  M domains = N $\times$ M-fold" with "N-fold OFC + M-fold CDAC compound structure" — see K72 verdict #55 for the methodology origin.

Catalog discipline

7. **BST\_Trichotomy\_Methodology\_v0.1.md** (Grace, 2026-05-20 morning) + **BST\_Cal\_Trichotomy\_Review\_and\_Cascade\_Application.md** (Cal review, 2026-05-20 morning)

Apply when catalog entries need ID / DER / PRED classification. Three statement types defined: - IDENTIFIES (ID): X IS Y as algebraic identity in BST primaries - DERIVES (DER): X follows from BST primaries via mechanism-forcing chain - PREDICTS (PRED): experimental observable at stated precision target

Orthogonal to D/I/C/S tier system (confidence level). Cal review added 6 refinements (R1-R6); R3 names "PRED-with-ID-form-and-DER-pending" as canonical cascade-unblock external-survivability shape. Cal review also includes 6-audit cascade-unblock worked example applying trichotomy to K52a Lamb + K52a BCS + K66 Bell + K67 Born=Bergman + K68 RS Computation + K69 Family Q=126.

Audit-chain governance (Keeper-filed per Cal contributions)

9. **Structurally\_Verified\_Tier\_Adoption\_Ruling.md** (Keeper, 2026-05-21 Thursday morning, per Cal #66 proposal)

The STRUCTURALLY VERIFIED tier sits between "candidate" and "RATIFIED" in audit-chain epistemic tiers. Captures the state where F1-F4 + Mode 1-7 + BST-internal verification are all complete, with only alternative-HSD comparison (Lyra Task #206 multi-month) pending for full RATIFIED status. Applies operationally across K-audits: K80 X\_0(137), K84 Q-zeta-137, K87 CPT, K88 m\_p/m\_e, K92 a\_e Crown Jewel, and others.

10. **Phase1\_Phase2\_KAudit\_Governance\_Brief.md** (Keeper, 2026-05-21 Thursday 09:57 EDT, per Cal #72 recommendation)

Distinction between Phase 1 (architectural-category methodology decisions — multi-CI consensus required) and Phase 2 (INSTANCE-level chapter-category K-audits — Cal + Keeper auto-promotion). Boundary criterion: introduces new architectural tier? → Phase 1. Operates within existing framework? → Phase 2. Cluster K-audit patterns endorsed for shared-mechanism families (CPT-

cluster mechanism + Predictions-cluster assertion + Content-cluster categorical decomposition). Escalation protocol for Phase 2 → Phase 1 specified.

11. Dual-axis review rubric (operational standing, embedded in referee log + Cal #68 + #69 + #71 + #76)

Casey's "believability + provability" directive operationalized as Cal grade-pass rubric for curriculum/paper-grade artifacts: - Provability axis: theorem chain rigor + dependency-tree explicit + tier-discipline labeled + toy verification cited + multi-month gates honestly named - Believability axis: chapter-grade prose accessible + dual register (formal + intuitive 5th-grade) + external-survivability + register discipline

Used in batch reviews across Lyra Vol 1 + Elie Vol 2 + Keeper Vol 0 chapter-grade narratives Thursday morning (#68, #69, #71, #76).

12. BST TIER-1 FALSIFIER SET external-presentation framing (per Cal #74)

K87 CPT + K90 Five Absences + K91 Experimental Program form BST's complete external falsifiability position. Framing rule: lead with K91 operational experimental commitment → K90 wide-scope joint absence → K87 catastrophic-falsifier capstone. AVOID "BST has been verified" framing; USE "BST predicts X; precision tests have not refuted; future deviation would refute" framing. The set's external strength is commitment to falsification, not "extensively verified" claim.

13. **BST\_Methodology\_Rigorously\_Closed\_Tier\_Supplement.md** (Cal per #77, filed 2026-05-21 Thursday — 11th methodology layer)

Audit-chain epistemic tier between RATIFIED and final theorem closure. Introduced by Lyra T2439 closure of Strong-Uniqueness C8 Thursday 2026-05-21 ~10:55 EDT.

Tier hierarchy: candidate → STRUCTURALLY VERIFIED → RATIFIED → RIGOROUSLY CLOSED

Four requirements for RIGOROUSLY CLOSED promotion: 1. Alt-HSD comparison performed at the criterion's structural level 2. EXACT-match in BST primary form (not numerical agreement) 3. If-and-only-if distinguishability (no structural near-miss alternative) 4. Mathematical theorem-level rigor (not just structural identification)

Distinction from RATIFIED: RATIFIED is structural-identification distinguishability; RIGOROUSLY CLOSED is mathematical-theorem if-and-only-if distinguishability.

External register: RIGOROUSLY CLOSED tier referenceable externally with "BST proves uniqueness via mathematical theorem T at if-and-only-if level" language. Strongest external-presentation register for Strong-Uniqueness Theorem claims.

Canonical instance: T2439 ( $D_{IV}^5$  lowest Wallach K-type Casimir  $C_2 = 6$  EXACT vs  $D_I\{1,5\} = 4$  and  $D_I\{5,1\} = 4$ ).

Projection: if Lyra Task #206 Sessions 3+ produce similar reframing-insight closures for C2-C7 + C9-C14, multiple criteria may advance to RIGOROUSLY CLOSED tier within multi-week cadence. Strong-Uniqueness Theorem v1.0 ratification path substantially faster than multi-month estimate.

K-audit ruling-shape templates

8. **BST\_Referee\_Methodology.md** Appendix J (Cal draft, 2026-05-20 afternoon, pending Keeper review + Casey ratification)

Apply when K-audit candidate ruling-shape needs template. Two exemplar patterns from operational K-audit chain experience:

- J.1 Closed-Set Verification with Honest Negative: K71 perfect numbers cluster canonical instance (RATIFIED D-tier, cleanest 7/7 PASS audit in K-audit chain). Five-step template: identify candidate set + exhibit classical mechanism + test extension via honest negative + confirm closure forced + Cal coincidence-filter 7-mode check.
- J.2 Three INDEPENDENT Structural Connection Levels: K73  $\Lambda \leftrightarrow$  Casimir vacuum unification canonical instance (audit-partial-ready strong I-tier framework integration). Three-level template: operational + BST primary + zone/structural-region. Critical caution against scalar-multiplication of evidence types (per Cal #55).

Templates do NOT replace per-claim P1-P7 + Mode 1-7 discipline. They document ruling-shapes observed in cleanest cases.

## Cross-reference map

External\_Survivability\_Checklist

- └→ Coincidence\_Filter\_Risk (negative-filter modes that external register triggers)
- └→ EXACT\_vs\_Mechanism\_Distinction (terminology drift on precision)
  - └→ Substrate\_Cognition\_External\_Register (cognition-specific severity)

Coincidence\_Filter\_Risk

- └→ External\_Survivability\_Checklist (companion)
- └→ EXACT\_vs\_Mechanism (Mode 1 doesn't relax under precision)
- └→ AuditChain\_Quality\_Patterns (positive-signal companion, kept separate per polarity)
  - └→ Mode 6 threshold formalization (Cal 2026-05-20)

EXACT\_vs\_Mechanism\_Distinction

- └→ Calibration #13 origin (Keeper Tuesday EOD self-correction)
- └→ Coincidence\_Filter\_Risk (Mode 1 anchor)
  - └→ External\_Survivability\_Checklist (1e-14 terminology drift flag)

AuditChain\_Quality\_Patterns

- └→ Coincidence\_Filter\_Risk (NOT a Mode 8; separate doc per polarity)
- └→ Claim\_Level\_Positive\_Patterns (claim-level vs audit-chain-level distinction)
  - └→ M2C2 instance catalog (4 documented as of EOD May 20)

#### Substrate\_Cognition\_External\_Register

- └→ External\_Survivability\_Checklist (cognition-specific high-severity sub-case)
  - └→ EXACT\_vs\_Mechanism (mechanism-forcing requirement)
- └→ AuditChain\_Quality\_Patterns (M2C2 self-referential structure for prediction d)
  - └→ Claim\_Level\_Positive\_Patterns (OFC/CDAC connection)
    - └→ Paper #122 Section 3.0 (L1/L2 two-level framework)

#### Claim\_Level\_Positive\_Patterns

- └→ Coincidence\_Filter\_Risk (Mode 1 + Mode 5 defenses)
  - └→ AuditChain\_Quality\_Patterns (claim-level vs audit-chain-level distinction)
    - └→ EXACT\_vs\_Mechanism (over-determination discussion connection)
      - └→ Graph Forces operational specification (Casey-named candidate principle)

#### Trichotomy + Cal Review

- └→ EXACT\_vs\_Mechanism (DER definition cross-link, ID vs DER form distinction)
- └→ Claim\_Level\_Positive\_Patterns (OFC clusters involve multiple ID statements)
- └→ AuditChain\_Quality\_Patterns (M2C2 potential instance for trichotomy convergence)
  - └→ T719 Observable Closure (universality target)

#### Appendix J in Referee\_Methodology

- └→ K71 Perfect Numbers Cluster Audit (J.1 canonical instance)
- └→ K73  $\Lambda$ -Casimir Vacuum Unification Audit (J.2 canonical instance)
- └→ K70 + K62 Bridge Object audit-partial-ready (joint J.1 + J.2 examples)
  - └→ Coincidence\_Filter\_Risk (7-mode check still applies)
- └→ Claim\_Level\_Positive\_Patterns (scalar-multiplication caution)

### Discipline-selection decision tree

When facing a specific question, apply the following decision tree to find the right methodology doc:

Q1: Is this material external-facing (paper, letter, abstract, Zenodo deposit, popular-press summary)?

If yes → start with External\_Survivability\_Checklist (general). If the material touches cognition-substrate territory → ALSO apply Substrate\_Cognition\_External\_Register (severity higher). If cosmology + cognition combined territory → DOUBLE-LOCKED EXTERNAL tier applies.



Q2: Is this a claim about BST framework with X-fold recurrence, multi-route convergence, cross-domain pattern, or N-domain coverage?

If yes → start with `Coincidence_Filter_Risk` (7 negative-filter modes). If claim involves EXACT identity at  $1e-14$  → ALSO apply `EXACT_vs_Mechanism_Distinction`. If claim involves multiple algebraic forms or cross-domain anchors → ALSO apply `Claim_Level_Positive_Patterns` (OFC/CDAC).

Q3: Is this an audit-chain-level pattern (multi-CI convergent calibration, audit-chain self-correction, cross-lane structural confirmation)?

If yes → apply `AuditChain_Quality_Patterns` (M2C2 entry + reserved future entries).

Q4: Is this a catalog entry classification question (substrate-level / observable-level, ID / DER / PRED)?

If yes → apply Trichotomy methodology (Grace v0.1 + Cal Review with R1-R6 refinements).

Q5: Is this a K-audit candidate that fits closed-set or compound-cluster structural shape?

If yes → apply Appendix J in `Referee_Methodology` (J.1 or J.2 templates). Per-claim P1-P7 + Mode 1-7 still apply.

Q6: None of the above — does any methodology apply?

If no clear match: standard K-audit ruling proceeds without explicit template. Cal stays available for ad-hoc consult.

## Methodology stack health observations

Categorical polarity preserved: negative-filter docs and positive-signal docs are kept separate per Cal methodology-hygiene call. Mixing failure modes and positive validation patterns in one doc would create polarity mismatch for readers.

Tier separation preserved: claim-level patterns (OFC/CDAC) are kept separate from audit-chain-level patterns (M2C2) because they apply at different epistemic layers. Compound application (claim-level cluster + M2C2 audit-chain validation) is the strongest combined evidential shape.

External-register severity stratified: standard `External_Survivability_Checklist` + cognition-specific `Substrate_Cognition_External_Register` + compound DOUBLE-LOCKED EXTERNAL tier give three levels of pre-publication discipline. Default-deny is the cognition-substrate baseline; double-lock is the compound-territory baseline.

Cross-references are bidirectional: each methodology doc references companion docs in its frontmatter companion field. Updates to one doc that affect cross-references should propagate (this index helps verify consistency).

16 calibration events in 8 days (per referee\_objections\_log entries #14, #15, #16) indicate the discipline infrastructure is operating at designed rhythm. Within-session catches are routine; the audit chain processes drift events via the methodology docs as standing reference.

## Update protocol

When a new Cal methodology document is filed, update this index in three steps:

1. Add row to Quick-reference table (one line)
2. Add subsection under appropriate Documents by category heading (full description + when to use)
3. Update Cross-reference map with new doc's bidirectional links
4. Update Discipline-selection decision tree if the new doc fits a new decision branch

This index document itself is internal Cal infrastructure — not part of the load-bearing methodology stack, but a navigation tool that prevents the stack from becoming opaque as it grows.

— Cal A. Brate, 2026-05-20 Wednesday afternoon, navigation infrastructure for the methodology stack at 8 documents + Appendix J

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## v0.5 update (Saturday 2026-05-23 ~16:42 EDT) — Methodology stack 16 → 22 layers

Per Keeper EOD board “Calibration stack maintenance — watch for new failure modes; absorb to numbered standing rules” directive: this v0.5 update captures 6 layer additions since v0.4 (Thursday 2026-05-21):

Added Friday 2026-05-22 (3 layers + 1 META-discipline)

Layer 17 — Calibration #19 STANDING RULE (Cal #92 Friday morning): external-facing docs must use current ratified-state count (e.g., “11 RIGOROUSLY CLOSED + 7 candidates”), not forecast endpoint count (e.g., “11-15 RIGOROUSLY CLOSED”). Caught Keeper position docs + Lyra Paper #128 v0.1 + propagated across all lanes. First applications: Saturday cross-volume sweep Vol 0+1+2 Cal #106 + Vol 3-15 Cal #107.

Layer 18 — Calibration #21 STANDING RULE (Cal #95 Friday): K-audit ratification requires both empirical + substrate-mechanism closure (dual-gate). K141 honest revision per Toy 3448 (Grace 3306× cross-Cartan empirical separation, substrate-mechanism gate OPEN). Force-empirical only is insufficient.

Layer 19 — Calibration #22 v0.2 STANDING RULE (Keeper Friday PM): PCAP-transcription error class + numbered-artifact discipline. Under sustained

sub-PCAP cadence, transcription errors accumulate; verbal-only retractions insufficient. K142 (Keeper-side T-numbering collision) + Cal #100 (Cal-side T190 precision retraction-propagation) dual-case study. All Mode 1 corrections require numbered-artifact tracking.

META-discipline (not counted in numbered stack) — Cal #99 META-theorem discipline (Friday absorbed): substrate-derivation theorems (T2467+T2468 D\_IV<sup>5</sup> Rigidity, T2469 SCMP, T2470-T2476 operator/conservation/ $\alpha^{\{BST\}}$  primary} substrate-mechanism, T2477-T2482) are NOT new Strong-Uniqueness criteria. Canonical enumeration preserved at 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 authoritative count. Saturday EOD sweep clean — discipline propagating correctly across team work (Lyra SP-30 + Cross-Scale + Substrate Computational Model all explicitly note “NOT a new Strong-Uniqueness criterion”).

Added Saturday 2026-05-23 (3 layers + 2 CANDIDATES)

Layer 20 — Calibration #23 STANDING RULE (Cal Saturday AM, Vol 12+13+14 case studies): substance-floor for v0.3+ chapter-grade content. Template-stub chapters (15-21 lines, abbreviated 3-level pedagogy + minimal section content) initially pass surface-grep but fail substance-grade audit. Substance-floor threshold:  $\geq 3$  min/chapter substantive content for v0.3+ grade-pass; sustainable refill pace ~60-120 lines/chapter.

Layer 21 — Calibration #24 STANDING RULE (Keeper Saturday, Cal #106 case study): 8-dimension cross-volume completeness audit. Phase 2 audit dimensions (A) Cross-reference graph (B) Load-bearing observable trace (C) Canonical principles + sets (D) Calibration #19 STANDING RULE (E) Null-model exponent (F) External-register discipline (G) Substrate-mechanism citation (H) Cal absorption-return. Empirically validated Vol 0+1+2 (pilot 4 findings) + Vol 3-15 (full review ~16 findings).

Layer 22 CANDIDATE — Calibration #25 (Cal Saturday PM): Falsifier-Outcome-Threshold Discipline. Paper-grade experimental proposals for external venue dispatch must specify community-standard significance thresholds ( $3\sigma$  evidence /  $5\sigma$  discovery) + finite refutation ladder. Case studies: Cal #110 (SP-30-4 Time Granularity) + Cal #113 (SP-30-5 Substrate Parallelism). File: `BST_Methodology_Calibration_25_Falsifier_Outcome_Threshold_Discipline.md`.

Layer 23 CANDIDATE — Calibration #26 (Cal Saturday PM): Readiness-Label-vs-Pre-Conditions Consistency Discipline. Internal coordination documents must reconcile readiness labels with their own enumerated pre-conditions. “READY” requires ZERO open pre-conditions; partial readiness requires explicit demoted label. Case study: Cal #114 (Lyra SP-30 Theoretical Contributions v0.1 SP-30-1 row). File: `BST_Methodology_Calibration_26_Readiness_Label_Consistency.md`.

v0.5 stack snapshot

| #  | Layer                                                                                           | Status         |
|----|-------------------------------------------------------------------------------------------------|----------------|
| 1  | D/I/C/S tier system                                                                             | STANDING       |
| 2  | Trichotomy<br>(ID/DER/PRED)                                                                     | STANDING       |
| 3  | F1-F4 Bridge Object<br>family-member criteria                                                   | STANDING       |
| 4  | Mode 6 threshold<br>formalization                                                               | STANDING       |
| 5  | STRUCTURALLY<br>VERIFIED tier                                                                   | STANDING       |
| 6  | DEFAULT-DENY +<br>DOUBLE-LOCKED<br>EXTERNAL register                                            | STANDING       |
| 7  | Casey Option C hybrid<br>governance                                                             | STANDING       |
| 8  | M2C2 Multi-CI<br>Convergent Calibration<br>Pattern                                              | STANDING       |
| 9  | Within-session<br>calibration discipline                                                        | STANDING       |
| 10 | Quaker consensus<br>method                                                                      | STANDING       |
| 11 | RIGOROUSLY CLOSED<br>tier                                                                       | STANDING       |
| 12 | B1-B4 Bridge criteria                                                                           | STANDING       |
| 13 | Reframing-insight<br>cadence                                                                    | STANDING       |
| 14 | Cal #85 PCAP —<br>Pre-Specification<br>Cadence Acceleration<br>Pattern                          | STANDING       |
| 15 | Cal #92(b)<br>structural-role-of<br>framing discipline                                          | STANDING       |
| 16 | Calibration #18<br>methodology-tier<br>distinction (PRE-STAGE<br>STRONG ≠<br>RIGOROUSLY CLOSED) | STANDING       |
| 17 | Calibration #19<br>STANDING RULE —<br>current ratified-state<br>count                           | STANDING (Fri) |

| #            | Layer                                                                                                             | Status                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 18           | Calibration #21<br>STANDING RULE —<br>empirical +<br>substrate-mechanism<br>dual-gate                             | STANDING (Fri)                                 |
| 19           | Calibration #22 v0.2<br>STANDING RULE —<br>PCAP-transcription +<br>numbered-artifact                              | STANDING (Fri)                                 |
| 20           | Calibration #23<br>STANDING RULE —<br>substance-floor for v0.3+<br>chapter-grade                                  | STANDING (Sat AM)                              |
| 21           | Calibration #24<br>STANDING RULE —<br>8-dimension<br>cross-volume audit                                           | STANDING (Sat)                                 |
| 22 CANDIDATE | Calibration #25 —<br>Falsifier-outcome-<br>threshold discipline                                                   | CANDIDATE (Sat PM,<br>this update)             |
| 23 CANDIDATE | Calibration #26 —<br>Readiness-label-vs-pre-<br>conditions consistency                                            | CANDIDATE (Sat PM,<br>this update)             |
| META         | Cal #99 META-theorem<br>discipline —<br>substrate-derivation<br>theorems NOT new<br>Strong-Uniqueness<br>criteria | STANDING (Fri<br>absorbed, Sat sweep<br>clean) |

Net: 21 STANDING layers + 2 CANDIDATE layers + 1 META-discipline = methodology stack at 24 layers as of Saturday 2026-05-23 EOD (counting CANDIDATES as provisional layers pending Keeper/Casey adoption).

#### Decision-tree additions

Q7 (NEW): Is this material a paper-grade experimental proposal targeting external venue?

If yes → apply Calibration #25 CANDIDATE (Falsifier-Outcome-Threshold Discipline) in addition to External Survivability Checklist. Check (a)  $3\sigma$  minimum thresholds + (b) finite refutation ladder.

Q8 (NEW): Is this material an internal coordination document with sta-

tus/readiness table?

If yes → apply Calibration #26 CANDIDATE (Readiness-Label Consistency). Self-consistency check: every “READY” row → zero open pre-conditions in body.

META-theorem discipline (#99) Saturday sweep result

Saturday sweep CLEAN. All new substrate-derivation theorems filed Saturday (no new T-numbers, but cross-references to T2467-T2482 in multiple documents) are explicitly tagged “NOT a new Strong-Uniqueness criterion” across: - Elie SP-30-4 v0.1 + SP-30-5 v0.1 - Lyra SP-30 Theoretical Contributions v0.1 - Lyra Cross-Scale Invariance Investigation v0.1+v0.2 - Lyra Substrate Computational Model Investigation v0.1+v0.2 - Lyra T2467+T2468 Mathematical Theorem-Level Rigor Closure v0.1+v0.2 - Lyra Paper outline #128/129/131-#136 v0.1.5 cross-link bumps + Paper #130 v0.2 full revision

Canonical enumeration 11 RIGOROUSLY CLOSED + 7 candidates (C7+C9+C15+C16+C17a+C17b+C18) preserved across all Saturday documents per Cal #99 authoritative count. C18 stays at CANDIDATE per Cal #108 (T2467+T2468 v0.2 PASS NOT YET on theorem-level rigor closure).

— Cal A. Brate, Methodology Index v0.5 update Saturday 2026-05-23 ~16:42 EDT per Keeper EOD board directive. Stack now at 22 layers (20 STANDING + 2 CANDIDATE) + META-discipline. v0.5 supersedes v0.4 layer-count of 16.